# Sudhanshu Agrawal

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## Education

**Stanford University** **Sept 2026 – June 2028 (anticipated)**
M.S. Computer Science

**University of California, Los Angeles** **Sept 2019 – June 2023**
B.S. Computer Science *(Magna cum laude)* B.S. Mathematics *(Cum laude)* **GPA: 3.92**

**Academic Research**

- UCLA Computer Science: generative modeling, advised by Professor Aditya Grover.
- UCLA Mathematics: mean-field games, advised by Professors Levon Nurbekyan and Samy Wu Fung.

## Work Experience

**Qualcomm** : ***ML Research Engineer*** **2023 – 2026** *(San Diego)*

- **LLM research:** invented algorithms and architectures to accelerate LLM generation via speculative decoding. Commercialized research, published 5 papers, lead and co-invented 12 inventions with several US/global patents pending.

**Qualcomm** : ***ML Engineering Intern*** **Summer 2022** *(San Diego)*

- **Novel *Python* profiler:** created a profiling tool for deep learning applications, capable of timing code function-by-function, line-by-line, and profiling multiprocessing applications.

**SonicJobs** : ***ML Engineering Intern*** **Summer 2021** *(Remote)*

- **Synthetic data-set generation:** created a pipeline to scalably generate over 100,000 webpage samples with varying contents and styles, solving the challenge of obtaining a diverse, labeled, computer-vision training dataset.

**Reliance Jio** : ***ML & Data Science Intern*** **Summer 2020** *(Remote)*

- **Predicting the properties of hydrocarbons:** developed predictive models leveraging lasso regression, ridge regression, support vector regression, recursive feature elimination, linear discriminant analysis, and 1-D convolutions.

## Publications

**Analyzing LLM Hidden States using Information Theory** **2026**

- Goel, Raghavv*, **Agrawal, Sudhanshu***, et al., "Surfing the Activation Wave: Analyzing the Information and Geometry of LLM Hidden States for Speculative Decoding." **NeurIPS 2026 AXIOM, PriGM, GDDL Workshops**.

**Speculative Decoding for Diffusion LLMs** **2026**

- **Agrawal, Sudhanshu**, et al., "Structuring The Future: Diffusion LLM Speculative Decoding via Calibrated Draft Graphs." **ICML 2026 Foundations of Deep Generative Models and Structured Probabilistic Inference Workshops**.

**Representations in AR-LLMs vs dLLMs** **2026**

- Goel, Raghavv*, Garrepalli, Risheek *, **Agrawal, Sudhanshu**, et al., "A Comparative Analysis of Layer-wise Representational Capacity in AR and Diffusion LLMs." **ICLR 2026 Workshop on Science for Deep Learning**.

**Efficient Speculative Decoding via Vocabulary Dimensionality Reduction** **2025**

- Goel, Raghavv, **Agrawal, Sudhanshu**, et al., "VOCABTRIM: Vocabulary Pruning for Efficient Speculative Decoding in LLMs." **ICML 2025 Workshop on Efficient Systems for Foundation Models**.

**Early Draft Stopping for Speculative Decoding of LLMs** **2024**

- **Agrawal, Sudhanshu**, Jeon, Wonseok, and Lee, Mingu, "AdaEDL: Early Draft Stopping for Speculative Decoding of Large Language Models." **NeurIPS 2024 Efficient Natural Language and Speech Processing Workshop.**

**Foundation Models for Offline Black-Box Optimization** **2023**

- Nguyen, Tung, **Agrawal, Sudhanshu**, and Grover, Aditya, "Expt: Synthetic pretraining for few-shot experimental design." **Advances in Neural Information Processing Systems 36 (2023): 45856-45869 (NeurIPS 2023)**.

**Efficiently Solving High Dimensional Non-local Mean Field Game Problems** **2022**

- **Agrawal, Sudhanshu***, Lee, Wonjun*, Fung, Samy W., Nurbekyan Levon, "Random features for high-dimensional nonlocal mean-field games." **Journal of Computational Physics 459 (2022): 111136**.